Finger Millet Production

Manual for Uganda



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Finger Millet Production Manual

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Acronyms

NARO	National Agricultural Research Organization
NaSARRI	National Semi-Arid Resources Research Institute
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics

Foreword

Finger millet is one of the major cereals grown in Uganda with wide adaptability ranging from sub humid agro-ecologies to semi-arid regions in the north and north eastern Uganda. it's mainly cultivated by small scale farmers in marginal land area without any input. Even under this condition finger millet yield stands at 1.1 tons /ha but the potential yield of finger millet under proper management is 3 to 5 tones/ha. With increasing effects of climate change where most parts of the country are becoming hotter and drier, finger millet may regain its importance as staple food crop. Finger millet has high protein content, balanced amino acid profile, and high levels of minerals particularly calcium, iron, zinc, phosphorus and insoluble dietary fiber as well as anti-oxidants. These adaptive and nutritional features combined with high yield potential make finger millet an important cereal crop that can effectively address the emerging challenges of climate change, water shortages, land degradation and food-related health issues. However, the production of finger millet in Uganda is largely constraint by number of factors including; poor agronomic practices, diseases (like finger millet blast), parasitic weed infestation, declining soil fertility, erratic rainfall and low yield potential of currently grown varieties/ land races as well as lack of information about the crop.

National Agricultural Research Organization through the Dry Land Cereals Research Program at National Semi Arid Resources Research Institute is responding to these challenges by developing new and better adapted varieties along with crop management practices. In this case NARO-NaSARRI has developed finger millet production manual which provides guideline on the different agronomic practices, pest and disease management, weed management and post-harvest handling of finger millet. The manual is useful for wide continuum of stakeholders including farmers, extension officers, researchers and other stakeholders involved in finger millet value chain in Uganda.

I thank the authors for their efforts in putting the information contained in this manual together and I do hope that you will find the content

of this document useful in enhancing production and productivity of finger millet in the country.

Dr. Ambrose Agona

Director General

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CHAPTER ONE

Introduction

Background

Finger millet is a seeded annual cereal which belongs to the grass family, Poaceae. The height of a mature plant ranges from 30-150 cm depending on the variety and environmental conditions where it's grown. In Uganda, among the cereal crops grown, finger millet ranks third to maize and sorghum in terms of total national production with an average yield of 1.6 metric tons per ha. The crop is grown throughout Uganda but mainly in the Eastern, Northern and Western regions by small-scale farmers and purely at subsistence level. Up to 65% of the country's finger millet is in the greater districts of Kabale, Hoima, Bushenyi, Masindi, Mbarara in western region; Iganga, Kamuli, Kumi, Soroti, Mbale, Tororo in eastern region; and Gulu, Kitgum, Lira, Apach and Arua in the northern region. Production of the crop is likely to expand into non-millet producing districts due to changes in climatic condition and ability of the crop to survive and perform well in areas with less moisture.

Finger millet is a staple diet of many subsistence farmers and rural workers in Uganda, and is an important source of energy and proteins. The crop has high levels of minerals (especially calcium, manganese, phosphorus, zinc and iron) and slowly digestible fibre compared with other cereals thus providing health benefits such as preventing constipation, anemia, aiding bone strength, and inducing production of sufficient breast milk in lactating mothers as well as food for people with diabetes. Good storage properties and pathogen resistance makes finger millet an ideal crop for poverty alleviation, and for food and nutritional security. In addition, the crop has the ability to tolerate stress and can be grown in marginal land without the use of inputs like fertilizers.

Despite its importance, finger millet productivity has remained low with on farm average yield of 700 kg/ha compared to its potential yield

of 5 tones/ha. This low yield is a result of many factors including low yielding varieties, poor agronomic practices, insect pest and diseases, declining soil fertility and poor post-harvest handling among others. Several efforts have been made to improve finger millet and new varieties have been released. Unfortunately, some of these improved varieties have never been accessed by farmers mainly because of lack of information about them and the accompanying agronomic practices. In addition, farmers also lack knowledge on the post-harvest handling processes of finger millet resulting into greater grain loss in terms of quality and quantity.

To improve finger millet productivity, there is a need to create awareness on the crop in terms of its nutritional importance and to avail farmers with knowledge on the recommended agronomic and post-harvest handling practices. This can be achieved by availing and sharing the information about the crop with different stakeholders like extension officers, researchers, policy regulators and farmers. This manual, therefore, will provide technical guidance on finger millet agronomy, pest and disease management, weed management and post-harvest handling processes. It will be of great use to both farmers and extension officers involved in finger millet production in Uganda.

Health benefit of finger millet

Finger millet is one of the most nutritious cereals. The grain is characterized by high concentration of carbohydrate, dietary fibre, phytochemicals and essential amino acids, presence of essential minerals as well as gluten free status. The rich nutritional properties of finger millet has several health benefits as discussed below:

Controls diabetes: Finger millet has high fibre content and low glycemic index that slows down the rate of digestion and thus helping in lowering the blood sugar levels. Therefore, regular consumption of finger millet will help reduce the risk of developing Diabetes Mellitus.

Improves bone health: Finger millet is good source of calcium and helps joining bones. Consumption of finger millet can help developing bones in growing children and maintaining the bone strength of adults. It is also instrumental in keeping ailments like osteoporosis away and can

also cut down the risk of fracture.

Reduces the risk of anemia: Finger millet is an excellent source of natural iron and thus a vital food for anemic patients and for those with low hemoglobin levels.

Anti-cancer potential: Finger millet is rich in antioxidants that is necessary to prevent excessive oxidation, which could otherwise cause cancer and ageing because of cell damage. Phenolic acids, flavonoids and tannins present in finger millet seed coats have very effective antioxidant properties. In general, it has been shown that people on millet-based diets have lower frequencies of esophageal cancer than those on wheat or maize-diets.

Helps in Digestion: High level of dietary fiber in this cereal aids in proper digestion, normal bowel movement and prevents constipation. Insoluble fibers present in finger millet assists movement of food through the intestines while the insoluble fibers retain water thus easing the passage of waste.

Increases Lactation: Lactating mothers are recommended to include finger millet in their daily diet in order to improve milk production. It also enriches the milk with all the essential minerals like iron, calcium and vital amino acids that is extremely important for the child as well as the mother

Improves the performance of the heart by reducing Cholesterol: Finger millet controls the cholesterol level by reducing plaque formation and preventing blockage of blood vessels thus cutting down the risk of stroke and other heart diseases. The key amino acids Lecithin and Methionine present in the cereal helps in lowering cholesterol by extracting and cutting out excess fat from the liver, and the amino acid Threonine prevents formation of fat in the liver.

Finger millet contains vital amino acids methionine and lysine which is lacking in most cereals. These amino acids help in formation and maintenance of collagen which is key in supporting the skin tissues making it less prone to wrinkles and sagging. Hence finger millet act as anti-aging food

CHAPTER TWO

FINGER MILLET CLIMATIC CONDITION AND AGRONOMIC PRACTICES

Climatic conditions for finger millet production

Finger millet grows best in an environment with medium rainfall ranging between 500-1000 mm with optimum of 900 mm, well distributed during the growing season without prolonged droughts. The crop can however tolerate rainfall as low as 130 mm and does not tolerate flooding. Finger millet grows best where average maximum temperatures exceed 27 °C and average minimum temperatures do not fall below 18 °C, but can grow in temperatures up to 35 °C. Dry weather is required for drying the grain at harvest as the crop is harvested at physiological maturity to avoid shattering on drying in the fields. Areas with low precipitation and low relative humidity during seed ripening and maturation are best for regeneration.

Soil requirements

Finger millet thrives best in soils with pH ranges of 4.5 to 8.0. The crop can grow on a wide variety of soils ranging from rich loam to poor shallow upland soil with good organic matter. Black soil with good drainage can also be considered for finger millet cultivation. However, heavy clay soil with poor drainage should be avoided.

Field selection and preparation

The crop requires a well-prepared seedbed because of its small seed size, and inability to stand weed competition. Choose a field which was not under millet during the previous two years to reduce the risk of having volunteer plants. The selected field must have good drainage and should not be water logged. Timely field preparation facilitates early planting, which ensures higher yield. Land preparation should ensure that all crop residues, crop volunteers and weeds are completely buried in the soil.

Ploughing towards dry season is advantageous to kill the weed seeds and hibernating insects and pathogens by exposing them to the heat. Initial ploughing should be carried out at optimum moisture range to get good tilth and should be avoided when moisture is in excess. It is recommended that the field is ploughed twice followed by harrowing to obtain a fine seedbed that will allow easy planting and seed germination and emergence. Ploughing can be done using an ox-plough or tractor (Fig. 1(a)). In areas where ploughing operations with ox-plough or tractor are difficult to carry out, effective digging and turning of soil, removing perennial weeds and land smoothening using simple tools like hoes (Fig. 1(b)) are recommended. Ultimately, fine seedbed ready for planting (Fig. 1(c)) must be obtained.



Fig. 1 (a): Tractor ploughing



Fig. 1 (b): Hand hoe ploughing



Fig. 1 (c): Fine seedbed

Planting time

The best time for planting is as soon as the first effective rains (when there is enough moisture in the soil for seed germination) have been received. The time when the first effective rains are received varies from place to place, and has become quite unpredictable with the current changes in the rainfall pattern. However, first season planting can be done between March – April while the second season can be done between July – August depending on the start of rain season.

Seed requirement

I. Seed rate

Seed rate of 4 kg/acre is required to obtain optimum plant population.

II. Seed treatment

Currently in Uganda, finger millet seeds are not treated with any chemical. However, given that some seeds may be contaminated with fungal propagules that may cause diseases such as finger millet blast, seed treatment is highly recommended. Finger millet seed can be treated with Trycyclozole at 1g/kg seed to kill disease propagules as they germinate in the soil around roots. In cases where bio fertilizers are available, seeds can also be treated with nitrogen fixing bacteria like Azospirillum brasilense and phosphorus solubilizing fungi like *Aspergillus awamori to boost plant in mobilizing, fixing and uptake of nutrients.*

Plant spacing

For line sowing, spacing should be 30 cm between rows and 10 cm from plant to plant within rows. Optimum plant population in finger millet ranges between 400,000 - 500,000 plants per ha. In less fertile soils, a spacing of 50 x 20 cm can be adopted and anything below or above that will reduce the crop yield. Maintenance of optimum plant population is an important prerequisite for getting higher yields under rain fed conditions.

Planting Depth

Seeds are sown to a depth of not more than 2.5 cm. Over burying of seed will lead to no germination given the size of millet seed.

Planting method

The following are the common methods of sowing finger millet:

Broad casting:

This is the common method used by most farmers where the seed is directly sowed across an area by scattering. However, it requires a lot of seed and is very likely to result in non-uniform germination.

Line sowing:

In this method the seeds are sowed in line using simple fabricated machines like forked rake spaced at 30 cm apart. If a machine is lacking, manual line sowing is done. Planting in straight line will make it easier to weed or apply fertilizers, herbicides, or insecticides. More importantly, optimum plant population is obtained. The following steps are followed during manual line sowing:

i. *Mark out the field using a marked string at 30cm between rows*

ii. *Place the pegs at each 30cm spot (Fig. 2(a)).*

iii. *Place the strings marked at every 10 cm for spacing between holes until the end of the field.*

iv. *Plant seeds at every 10 cm mark along the string (Fig. 2(b)).*

The limitation with manual line sowing is that, it is tiresome and labour demanding especially in commercial fields and as such its use has been limited only to experimental fields in research stations but has not been adapted by farmers.



Fig. 2 (a): Placing guide sticks and lines for line sowing



Fig. 2 (b): Farmers sowing finger millet seeds along line



Fig. 2 (c): Planting using forked rake of 30 cm apart for drilling method.

Drilling in rows:

In this method, furrows 30 cm apart are created with help of a stick or forked rake (Fig. 2(c)) and the finger millet seeds are drilled in the furrow and later covered.

Transplanting the seedling:

Transplanting is a process of raising seedling in nursery beds and transplanting them into the main field. Leveling and watering of bed is required during transplanting. Seedling should be transplanted into the field when they are 4 weeks old.

Fertilizer use in finger millet production

Finger millets respond well to fertilizer application specifically to nitrogen and phosphorus with a blanket rate of 60 kg N, 30 kg P & 30 kg K. When correct amount and proper placement of fertilizer is used, the yield of finger millet increases three times. In finger millet, microdosing technique of fertilizer application is recommended. Microdosing is application of 1/3 of recommended fertilizer to a sowing hole or beside a plant at an appropriate time for more efficient use of the fertilizer and available moisture. The fertilizer is placed in the root zone of finger millet to facilitate efficient uptake.

Three methods of microdosing technique of fertilizer application are currently in use i.e.;

1. Planting holes are dug, then fertilizer and seeds are placed into them and covered

2. Fertilizer is placed 5 cm away from growing plantlets, to avoid the burning effect of fertilizer on the plantlets.

3. Appropriate amount of seed and fertilizer is mixed prior to seeding in moist soil conditions. Seeding under dry soil conditions leads to damage to the seeds or seedlings.

The microdosing technique of fertilizer application to be used depends on the sowing method of finger millet as indicated below:

Seeding by drilling: Suitable amount of fertilizer and seed is mixed immediately before drilling in moist soil conditions.

Spot sowing: Fertilizer is placed in the seeding hole.

Broadcasting: Seed is mixed with fertilizer and then immediately broadcast and worked into moist soil by pulling twigs over it.

Transplanting finger millet: Planting holes are dug then appropriate fertilizer is immediately placed in holes followed by transplanting of the seedlings.

Where rainfall is high especially around the Lake Victoria basin, delay of fertilizer application for 30 - 40 days after sowing is necessary. Under this system, the correct amount of fertilizer is spot placed 5 cm away from the target plantlet. Fertilizer can be applied in two to three equal doses starting at sowing, 30 days after sowing and at booting depending on moisture availability.

Use of chemical fertilizer requires technical expertise along with soil testing which is costly for most finger millet growers. However, use of organic manure (Fig. 3) is highly recommended in finger millet production and 5 - 10 ton of farm yard manure or well decomposed cow dung can be applied in the field one month before sowing.



Fig. 3: Organic manure ready for application

Thinning and gap filling

Irrespective of the method of sowing used except for transplanting, after crop establishment (about 20 - 27 days of sowing and plants are about 30 cm tall), thinning operation (Fig. 4) to remove excess seedlings must be done. Thinning is done 1 plant per hill. Similarly, where the plant population is not uniform, gap filling with 20 - 25 day old seedlings should be done. A plant population that is higher than optimal will result in tall and thin plants that will produce very small heads, resulting in reduced yields. On the other hand, a lower than optimal plant population will result in less heads, resulting in reduced yields.



Fig. 4: A woman thinning a finger millet field

Weeding

Weeds normally compete with finger millet for water, nutrients and light thus affecting plant growth and resulting into poor yield. Timely weeding is important, since finger millet seedlings are slow growing and require a weed free environment for the first 45 days to develop vigorous plants. Therefore, efficient weed control at the pre- and early post emergence stages is essential.

The number of weeding operations depend on the weed intensity but weeding twice is recommended to achieve optimum plant growth. The crop is mostly hand weeded using simple hoes (Fig. 5 (a)) but mechanical planting in (Fig. 5 (b)) rows makes weeding easier compared to broadcasting.



Fig. 5 (a): A farmer weeding a finger millet field



Fig. 5 (b): Nicely weeded field

Irrigation

Finger millet production in Uganda is purely rain fed. But if water stress occurs at tillering and flowering stages, it can cause significant yield loss even up to 100% loss depending on the length of drought. Therefore, irrigation is necessary (Fig. 6) for good plant growth along with good yield.



Fig. 6: Irrigation pipes running through a crop field

CHAPTER THREE

MAJOR PESTS AND DISEASES OF FINGER MILLET, AND THEIR MANAGEMENT

DISEASES OF FINGER MILLET AND THEIR MANAGEMENT

Finger millet blast

Blast caused by the fungus *Pyricularia grisea* is the major problem in Uganda. The disease affects finger millet at all stages of growth and causes yield losses ranging from 10% to 80% in Uganda depending on the variety susceptibility.

Symptoms:

Finger millet blast is characterized by the appearance of lesions on the leaves (Fig. 7 (a)), nodes (Fig. 7 (b)) and heads (Fig. 7 (C)). On the leaves, lesions are typically spindle-shaped, wide in the center and pointed toward either end. Large lesions usually develop a greyish center, with a brown margin on older lesions. Under blast disease-conducive conditions, lesions on the leaves of susceptible lines expand rapidly and tend to coalesce, leading to complete drying of infected leaves.



Fig. 7 (a): Finger millet leaves with multiple blast lesions



Fig. 7 (b): Blast fungus attack on nodes of plants causing neck rot



Fig. 7 (c): Finger millet heads affected by blast

Resistant plants may develop minute brown specks, indicative of a hypersensitive reaction. When the area between the leaf blade and leaf sheath (leaf collar) is infected, the collar turns brown and dies. This is called collar rot, which leads to death of the whole leaf above it.

Besides attacking the leaves, the fungus may also attack the neck/node causing neck rot (Fig. 7 (b)). When a node is infected, all parts above the infected node may die. When this occurs, yield losses may be great because grain formation is inhibited and/or formed grains may be shriveled. In such cases yield losses may be as high as 90%.

Management:

This can be achieved through cultural methods, use of resistant varieties and also chemical control application. These methods work best when applied in combination other than as a single method.

Cultural control methods:

These include, use of disease free seeds, time of planting, plant density, weed management and proper rotation as discussed below;

Use of disease free seeds: Majority of farmers who grow finger millet use their own saved seed which at times may be contaminated with fungal spores thus acting as primary source of inoculum. It is important to avoid contaminated seed by selecting ear heads that are free of disease and treating seed with a fungicide to kill disease propagules. Where possible, farmers can purchase certified seed from reputable seed dealers.

Early Planting time: Sowing seeds at the onset of the rainy season helps to avoid early infection of seedlings. This is because, by the time enough inoculum accumulates in the environment, the plants will have passed the sensitive periods for blast infection, which is between seed germination and plant tillering.

Soil fertility management: Excessive use of fertilizer is discouraged as it increases the incidence of blast. Heavy doses of nitrogen fertilizers cause lush crop growth rendering it susceptible to blast attack. A combination of organic and microdosing of inorganic fertilizer at ¼ of the recommended rate is both economical and can minimize blast infection. If animal manure is available, it should be used instead of nitrogen fertilizers. The rate of 5-10 t /ha animal manure is recommended. Nitrogen should be applied in small increments and not as one big dose.

Seeding rates: High plant density, coming from high seeding rate, enhances blast development and establishment due to dew accumulation on leaves. Seeding is therefore recommended to be done in rows and at the recommended spacing of 30×10 cm or 50x20 cm so as to allow an open space between the rows. This permits aeration and reduces dew accumulation.

Weed management: Finger millet must be kept weed free as weeds compete with the crop, weaken it thus facilitating easy infection with blast. A dense population of weeds creates a good micro-environment for development of blast due to increased humidity around the crop plants. Proper weeding to eliminate alternate blast pathogen host weeds will reduce disease levels.

Propagule dissemination control: Avoid working in the field when it is raining or when there is a lot of dew. Inoculum in the form of conidia sticks to clothes when they are wet and can be moved to other finger millet fields where they can initiate infection. Avoid use of same field tools in blast and non-blast infected fields to prevent blast propagule dissemination. It is advised to wash all field tools after using them in a blast infected field.

Practice crop rotation: Crop rotation is aimed at including in the cropping system, crops that are not infected by finger millet blast and at such intervals that reduce disease propagules in the field.

Intercropping: Finger millet can be intercropped with cowpea, groundnuts and pigeonpea so as to reduce disease propagules in the field

Field hygiene: Finger millet straw from diseased plants should either be ploughed deep under so that blast propagules are unable to germinate or it should be collected and burnt. Avoid threshing or winnowing finger millet from diseased plants in the field in order to reduce chances of disease propagules being blown long distances that can infect crops in other fields.

Grow resistant/tolerant varieties: Farmers should plant varieties that are resistant or tolerant to finger millet blast, e.g. the newly released varieties (NAROMIL 1 - 5, and Seremi 2).

Chemical control method:

Systemic fungicides such as pyroquilon and tricyclazone are possible chemicals for controlling the disease. Benlate has also been tried on finger millet and was found to reduce leaf blast by 28% on young millet and head blast by 38%. Finger millet seed can be treated with Trycyclozole at 1g/kg seed to kill disease propagules as they germinate

in the soil around roots. Such treatment eliminates disease infection through roots. Chemical blast control, though effective to a reasonable level, is a very expensive option for the resource poor farmer. Cultural control methods are the best options.

Downy mildew/Crazy top *(sclerophthora macrospora)*

Downy mildew in finger millet is known as crazy top downy mildew because of appearance of bushy and leafy growth in place of fingers or grains. The white cottony growth, characteristic of many other downy mildews, is generally not seen. The disease converts the heads partially or entirely into green narrow leafy structures causing complete sterility (Fig. 8 (a)). The proliferation starts at the basal spikelet and gradually the whole ear gives a bush-like appearance called 'green ear' symptom.



Fig. 8 (a): Bushy and leafy growth on millet head cause by mildew

The causal organisms *(sclerophthora macrospora)* survive as oospores in host tissues and soil. Primary infections usually occur in soil while secondary infections are transmitted through conidia fungi and sporangia.

Symptoms:

The symptoms are visible as either systemic or localized infection. Systemically infected seedlings are pale yellow or have light-color streaking on the leaf, chlorotic and stunted and may cause premature plant death. First symptoms are visible on the lower part of the leaf

blade, which later progress upwards. In cool, humid weather the lower surfaces of chlorotic leaves become covered by a white downy growth (Fig. 8 (b)) consisting of conidia and conidiophores of the pathogen.



Fig. 8 (b): White downy growth manifests itself in cool, humid weather

The leaves emerging from the whorl subsequently exhibit parallel stripes of vivid green and white tissue (Fig. 8 (c)). The infected striped areas die, turn brown, and disintegrate, resulting in a shredded appearance of the leaf.



Fig. 8 (c): Parallel stripes and browning caused by downy mildew

Management:

Growing disease-resistant cultivars is the most economical and feasible method of controlling downy mildew. Other measures include seed treatment with appropriate chemicals, providing good drainage in the

low lands, proper crop rotation methods, roguing of infected plants and elimination of wild grasses and related wild hosts to reduce the disease incidence.

Banded sheath blight *(Rhizoctonia solani)*

Banded sheath blight is a common disease among small millets. It is characterized by oval to irregular light grey to dark brown lesions on the lower leaf and leaf sheath. The symptoms produced on every part of the plant, give a very characteristic banded appearance, due to which the disease has been named as banded blight.

Symptoms:

Symptoms on millet-stalk can be easily observed on finger millets. The centre of the lesion subsequently turns white with narrow reddish brown margins. Later the spots get distributed irregularly on leaf lamina Fig. 9).



Fig. 9: leaf lamina infected with leaf branded sheath blight

Under favorable conditions, lesions enlarge rapidly and coalesce to cover large portions of the sheath and leaf lamina. At this stage, symptoms are characterized by a series of copper color bands across the leaves giving a banded appearance. In severe cases, symptoms appear on peduncles, fingers and glumes as irregular to oval, dark brown to purplish necrotic lesions. The mycelia growth along with sclerotia can be observed on and around the lesions.

Management:

Clean cultivation, draining out of excess water and removal of grass weeds on bunds can prevent the disease. Spraying with propiconazole (1 ml/l of water) is highly effective.

Small millet smut *(Melanopsichium eleusinis)*

Symptoms:

The affected ovaries are transformed into velvety greenish gall-like bodies (Fig. 10 (a)) which are several times bigger in size than the normal healthy grains. These infected grains gradually turn pinkish green (Fig. 10 (b)) and finally to dirty black on drying.

Management:

- Two sprays, the first with Difolatan at panicle initiation followed by a second with mancozeb at flowering can reduce disease incidence.
- The disease can also be controlled by adopting resistant cultivars, cultural practices like delayed sowing and seed treatment with Carboxinor Carbendazim @ 2 g per kg seed.



Fig. 10 (a): Finger millet head with swollen grains



Fig. 10 (b): Infected grains turn pinkish green before becoming dirty black

Small millet rust *(Uromyces eragrostidis)*

Symptoms:

Symptoms appear as minute to small, dark brown, broken pustules linearly arranged on the upper surface of the top leaves (Fig. 11). The disease is more on upper leaves compared to lower and middle leaves.



Fig. 11: Early symptom of finger millet rust

Uromyce setariae-italicae produces light yellow and black teleutospores. P. substriata develops small brown, oval spots on upper surfaces of the leaf. Telia are produced on the lower surface of leaf.

Management:

Growing of resistant varieties, planting clean seed, practicing crop rotation and weed control

Leaf spot *(Cercospora eleusine)*

Symptoms:

On finger millet the symptoms are usually first observed on the older leaves then spreading to the younger leaves. Initial symptoms appear as reddish-brown specks with yellow halo (Fig. 12). Later several such specks coalesce to form large lesions showing a burnt appearance. During rains the fungus sporulates producing grayish white growth at the centre of the spot.

Management:

- Growing of resistant varieties.
- Planting clean seed,
- practicing crop rotation with non-host plants especially legumes
- Proper weed management to reduce wild hosts.



Fig. 12: Leaf spot infection on plant

PESTS OF FINGER MILLET AND THEIR MANAGEMENT

Finger millet suffers from relatively fewer pests of economic importance both in the field and storage. The most common pests of finger millet include army worm (African army worm and fall army worm), cutworm, stem borer, leaf aphid, grasshoppers, grey weevil, shoofly and ear caterpillars as describe below.

Armyworms and cut worms

They appear during the early stages and continue up to harvest. The caterpillars Fig. (13 (b)) cut seedlings at the base, making the crop to appear as if grazed by domestic animals. They are active during night and hide under stones and clods during the day. In later stages of plant growth, these insects act as defoliators (13 (c)).



Fig. 13 (a): Adult moth



Fig. 13 (b): Armyworm larvae



Fig. 13 (c): Armyworms damage to plant

Management:

When the symptoms are noticed, the crop should be sprayed with contact pesticides like:

Rocket 44 EC (cypermethrin 4% + Profenos 40%) at 50 ml/ 15 litres water.

Cypermethrin 10 EC at 50 ml/ 15litre water sprayer

Leaf aphid

It occurs throughout the crop growing period. The nymphs and adults suck the sap from tender leaves and stem (Fig. 14 a and b). They can cause serious damage to seedlings up to 30 days.



Fig. 14 (a): Early stage of attack by leaf aphids



Fig. 14 (b): Aphids feeding on finger millet plant

Management:

- Planting resistant varieties of finger millet
- Practice crop rotation with non-host plant
- Spraying finger millet seedling with imidachloropid 17.8% EC (90-120ml/acre or 23- 30mls/15 litre of water

Stem borers

These borers attack seedling plants from 4 weeks of age through grain maturity. Early instar larvae enter the leaf whorl and feed on soft tissues; affected leaves show pinhole damage after they unfold. Later, larvae bore into stems (Fig. 15), forming frass-filled tunnels. Drying or wilting of the central shoot or growing point during the vegetative stage results in the dead heart condition. Side tillers with chaffy spikelets

are then produced in response to damage. Peduncles are damaged by late attacks, and many plants lodge, while those that remain standing produce white chaffy panicles (commonly known as 'white ear head').

Management:

- Crop residues or stubble should be ploughed down or burnt to destroy the hibernating larvae
- Plant resistant varieties of finger millet
- Remove plants with dead hearts and burry
- Set up light traps until midnight to attract and kill the stem borer moths
- Practice crop rotation with non-host plant
- Spraying finger millet seedling with imidachloropid 17.8% EC (90-120ml/acre or 23- 30mls/15 litre of water
- Spray the crop with Dimethioate (0.05%) or Phosphamidon (0.05%) or Monocrotophos (0.04%)



Fig. 15: A stem borer inside a plant stem

Ear head caterpillar

Ear caterpillars appear at dough stage on ears and persist till harvest. The larvae (*Fig. 16a*) destroy the grain in the head by eating the grains (*Fig. 16b*). They produce webs of silken thread that remain on and inside the head. Heavily infested heads may be covered with webbing (*Fig. 16c*). This further attracts saprophytic fungi.

Management

- Collect and burn stubbles and chaffy earheads, and feed the stalks to cattle before the onset of rains. This will prevent carryover of hibernating pest.
- Adopt synchronous and timely/early sowings of cultivars with similar maturity to reduce the damage of ear carterpillar
- Practice crop rotation with non-host plants like groundnuts, cotton and sunflower
- At milky stage when 1 to 2 bugs per panicle and head caterpillars (2 – 3 larvae per panicle), are noticed the crop may be sprayed at the milk stage with Cypermethrin 25% EC @ 0.5 ml/liter.

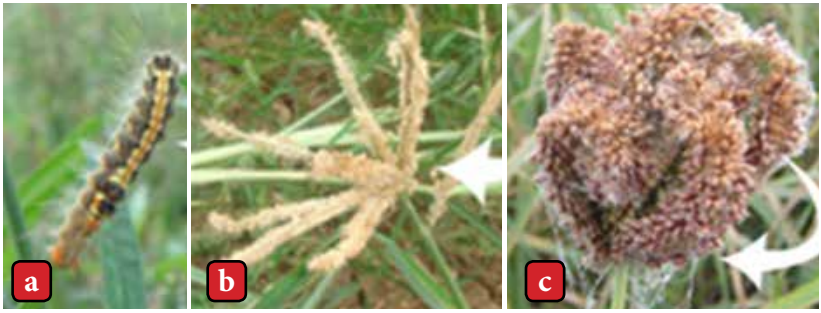


Fig. 16: Ear head caterpillar and its damage on finger millet

Fall army worm

The larvae scrap the chlorophyll which leads to white elongated patches. As larvae continue to feed, it causes windows on leaves because of voracious feeding and faecal pellets in whorls. Later larval instar enters in to shoot and feed inside and faecal pellets noticed on infested shoot. At flowering stage, larvae damage ear head of finger millet and faecal pellets are visible on finger

Management

- Plant immediately the onset of the rain, when the occurrence of the pest in the area is still low
- Deep plough the soil to expose insect stages hiding underground to birds and other natural enemies

- Use good quality seeds to increase plant vigour and potentially reduce damage
- Use millet varieties showing resistance/tolerance.
- Avoid staggered planting (planting at different dates in the same field), as this provides a continuous source of food for the pest
- Ensure optimum use of fertilizer for strong millet plants able to withstand or compensate for pest damage and apply at the right time.
- Intercropping with legumes helps to reduce pest pressure on both the millet and legume crops and supports natural enemies
- On small-scale farms, handpick and destroy the egg masses and larvae.
- If available, spray young caterpillars with neem-based products before they enter the whorl
- Use microbial biopesticides based on bacteria, fungi and viruses if available

CHAPTER FOUR

WEEDS AND WEED MANAGEMENT IN FINGER MILLET PRODUCTION

Major weeds and their effect on finger millet production

Weeds are a major obstacle in increasing the productivity of finger millet crops in Uganda especially during rainy season. Several weed species including grasses (especially wild relatives of finger millet like *Eleusine indica* and *E. africana* (Fig. 17 (a)) which are hard to distinguish from finger millet in vegetative stages), Sedges (e g *Cyperus rotundus* (Fig. 17 (b) and Nutsedge Cyperus (Fig. 17(c)), Johnson grass (Fig. 17 (d)), broad leaved weeds (e.g. *poison ivy* (Fig 17(e) and wandering jew (Fig 17 (f)), do affect millet production in the following ways:

- Competition for soil moisture - water is one of the limiting factors in finger millet production. Weeds often require more water than crops and are more efficient at capturing available soil moisture by developing deeper roots that spread wider than the crop roots.
- Competition for nutrients - Weeds are better competitors for nutrients than crops because of their more efficient root system and fast growth. Weeds compete for limited supplies of nitrogen, phosphorus and potassium in the soil thus depriving the crop plants of essential nutrients resulting in weak plants and poor yield.
- Competition for light - most weeds tends to grow faster than finger millet thus developing large canopies that easily intercept light at the expense of the crop.
- Some weeds are allelopathic in nature i.e. they produce certain chemicals or hormones that interfere with the growth of finger millet. For example, the nut sedges (Fig. 17 (c)) and Johnson grass (Fig. 17 (d)) are known to produce allelopathic chemicals that inhibit growth of finger millet
- Some weeds are parasitic in nature e.g. striga weed
- Weeds serves as alternate hosts for crop pests and diseases e.g. *Eleusine indica* is a common weed that serves as an alternate host

to finger millet blast.

- Weeds also affect the crop quality by producing a large number of seeds, which contaminate finger millet grain.
- Some weeds are poisonous to humans e.g. Poison ivy *(Toxicodendron radicans)* (Fig. 17 (e)) causes severe skin irritation (dermatitis) in humans.



Fig. 17 (a): Eleusine indica



Fig. 17 (b): Cyperus rotundus



Fig. 17 (c): Nutsedge Cyperus



Fig. 17 (d): Johnson grass



Fig. 17 (e): Poison ivy



Fig. 17 (f): Wandering jew weed

General weed control methods in finger millet

Weed management requires an integrated strategy rather than single approaches and it starts at the time of land preparation. In any crop production practice, weed control is limited to five options i.e. preventative, cultural, biological, mechanical, and chemical control. Choice of a particular method or methods depends upon the weed spectrum, crop rotation sequence, and other factors.

Preventive weed control: These are practices that will limit introduction or spread of weeds in the field. These include:

- Planting clean, weed-free finger millet seed.
- Ensuring that all equipment used to plant finger millet is free of weed seeds.
- Controlling weeds along ditch banks, roadsides, and field margins to prevent weed seed from entering fields.

Cultural weed control method: These are practices that allow the crop to become established without experiencing any negative effects of weed interference. These include:

Proper crop variety selection: Different genotypes of the same crop possess different traits that make them vary in their abilities to compete against weeds. These traits are typically those related to faster seedling emergence, quick canopy establishment and higher growth rates in the early stages. Use of these genotypes can therefore reduce the need for direct weed control measures (e.g. herbicides or cultivation)

Timely planting: This will ensure that crop emerge first giving it a competitive advantage for resources against the weed

Plant density: A good crop cover by adopting right spacing between rows and within the row will smother weed growth. Sowing finger millet by drilling eliminates spaces where weeds can grow within the plant row. The only available space for weed growth is between the rows where weeds can be removed by hand hoe in small farms.

Intercropping: Intercropping increases the use of natural resources by the canopy and, compared to sole crops, often compete better with weeds for light, water and nutrients. For example, compared to sole cropping, a finger millet cowpea intercrop sown in a two row-by-one row layout decreased relative soil cover of weeds.

Crop rotation: This is a practice of growing different crops in a sequence. When different crops are grown, different cultural practices are used and as such, cuts the life cycle of weeds.

Mechanical /Physical weed control: In the physical weed control method, weeds are removed from the field or are frequently disturbed to cause their death or inability to produce seed. There are several ways this can be done and these include;

- Turning the soil well during cultivation so as to cover all weeds to deny them re-growth.
- Slashing down all weeds so as to deplete their food reserves and cause their death. This can be done between the finger millet rows.
- Removing weeds by hand especially those growing very near the crop plants. This is possible where there is spot growth of weeds and is common for finger millet.
- Using of hand hoe, which is a very common practice by resource poor farmers. It is the best way of removing weeds between the rows of finger millet. Any weeds found within the rows are removed by hand.

Chemical weed control: Chemical control or herbicide application is advantageous in case of non-availability of labour and/or very high cost of labour for manual weeding. Use of correct herbicides will provide a completely weed-free situation to the crop whereas manual or mechanical weeding can be practiced only after the emergence of weeds. However, chemical weed control options are limited for finger millet production. Use of chemicals (herbicides) to control weeds should be based on their cost effectiveness compared to cultural and mechanical control measures. Herbicide application requires some knowledge about the use of such chemicals.

Striga weed in finger millet and its management

Striga also called witch weed is characterized by bright-green stems and leaves and small, brightly colored and attractive flowers (Fig. 18 (a) and (b)). The flowers are pink, red, white or yellow depending on the species.



Fig. 18 (a): Striga



Fig. 18 (b): A striga infested field

Economic importance of striga

Striga is a major biotic constraint in agriculture and causes considerable crop damage (70 to 100% yield loss) in cereals around the semi-arid tropics. The current increase in striga incidence is attributed to poor soil fertility and structure, moisture stress, intensification of land use through continuous cultivation and expansion of cereal production. As a result, small holder farmers are the most affected by striga because they have limited ways and means of control. Striga causes crop damage when it is still under-ground through competition for nutrients and through physiological interactions, and metabolic processes.

Symptoms of striga infestation are similar to those caused by insect pests, pathogens or environmental factors like nutrient deficiency or drought. The symptoms may include: stunted growth, yellowing, scotching of leaves, wilting and complete death of affected crop plant. The ultimate result of striga damage is a reduction in yield. The extent of yield loss is related to the incidence and severity of attack, the host's susceptibility to Striga, environmental factors (edaphic and climatic), cropping systems, local cultural practices, farmers' skills on the ecology, and farming practices.

Management of striga

Mechanical method:

Hand-weeding/hand-pulling is the most common control measure used by the small-scale farmers. Uprooting should be done before the striga plant starts flowering. The striga plant should be hand uprooted, collected and burnt so that seeds are not produced and shed from the drying plants. Avoid feeding livestock with witch weed plants as the seeds pass through the gut and are distributed in the manure.

Cultural method:

Use of resistant crop cultivars: It's the most economically feasible and environmentally friendly means of Striga control. Most of the released finger millet varieties are tolerant to striga and farmers should adopt them.

Practice rotation with trap crops: Rotation with non-host crops interrupts further production of Striga and leads to decline in the seed population in the soil. When trap crops are used in the rotation, they induce the germination of Striga but are not parasitized, leading to starvation and death of striga. They include cotton, cowpea, jute, soya bean, pigeon pea, chickpea, kenaf and groundnuts.

Rotation with catch crops: Catch crops are planted to stimulate a high percentage of the parasite seeds to germinate but are destroyed or harvested before the parasite can reproduce. For example, a thick planting of sudan grass at 20 - 25 kg seed per hectare should be sown and either ploughed in or harvested for forage at 6 - 8 weeks before Striga set seeds. The main crop could then be planted during the main rains.

Intercropping: Where intercropping is practiced, cereals should be intercropped with legumes in order to reduce the impact of striga.

Fertilizer application: Since striga is more favored in less fertile soil, good soil management practices like use of crop residues, organic manure or inorganic soil amendments will increase suppressiveness to striga species. For example, nitrogen fertilizer has been demonstrated to suppress striga growth

Chemical control methods:

Germination stimulants: Certain chemicals such as ethylene, ethephon, strigol and strigol analogues can induce germination of Striga seeds in the absence of a suitable host and therefore reduce seed reserves in the soil. Ethylene can reduce S. asiatica, however, *S. hermonthica* may not be well controlled by ethylene under field conditions in eastern Africa.

Herbicides: Many herbicides are useful in preventing the build-up of Striga seeds in the soil but may not prevent the damage done by Striga plants before emergence. Currently in Uganda there is no proper herbicide use in controlling striga in millet. However, some chemicals have been investigated for efficacy in controlling Striga in maize eg Dicamba and Imazapyr.

Integrated Striga management:

No single management method is effective against striga. Therefore, an integrated approach that combines the above control methods offer the best possibility for reducing the impact of striga at farm level.

CHAPTER FIVE

FINGER MILLET POST-HARVEST OPERATIONS

Finger Millet Pre-Harvest Operations

One of the important pre-harvest considerations that ensures produce quality is uniformity in the crop maturity. If ripening is not uniform (Fig. 19) in the field, harvesting should be done selectively, picking out the ripe heads that may start shattering while leaving the unripe heads for the next round of harvesting.



Fig. 19: Finger millet field with both ripe and unripe heads

Harvesting Finger Millet

The best stage to harvest the crop is when it has attained physiological maturity (Fig. 20). The crop is physiologically mature when leaves turn yellowish and begin to dry naturally, and the ears becomes brown(Fig.20). The grains are hard and firm. The longer the crop is kept in the field after physiological maturity, the greater the losses due to birds, insects, rodents, mold and bad weather. Harvesting is done using sharp hand sickles or knife. The ears are cut with about 2 cm of stalk.

In some cases, the harvested ears are then kept in a pile for one to two days under shade without drying to ripen the grain further and to give

the desirable taste; after which they are sun-dried.

Before harvesting is done, however, proper planning is essential. This may include making sure that all the necessary equipment is available, knowing where the drying will be done and how the grain will be stored. Holes in sacks should be mended and make sure stores are in good repair and that residues of previous harvest are removed.

Harvesting is usually desirable on a sunny day (Fig. 20). Make sure the crop is placed on a hygienic mat, tarpaulin or sacks. This prevents contact with the soil which can lead to moisture uptake, staining of the crop, and fungal infection that can lead to mycotoxin production.



Fig. 20: Harvesting of mature finger millet field

Late harvesting should be avoided as this may result into crop losses caused by several factors including shattering of seeds in field, infestation by pathogens, insect pests and damage by birds, other wild animals and losses due to theft. Late harvesting may delay planting of subsequent crops.

Transportation

Transportation of finger millet starts immediately after harvesting within the farm. The harvest should be transported to the homestead as soon as possible. Such transport is typically by head load, barrow, bicycle or ox drawn carts. It is important to make sure the crop is transported in clean dry and sealed containers.

Drying

Drying should preferably be conducted at homestead other than in the field. The harvested heads are spread directly under sunshine on cemented drying floor (Fig. 21 (a)), a tarpaulin (Fig. 21 (c)), layer of sacks or woven mats (Fig. 21 (c)) to attain moisture content of 12%. The crop should never be placed in direct contact with the soil, and should be kept away from farm animals, otherwise the grain may be damaged or eaten. This may be done by tethering animals or fencing in the area where grain is drying.

Steps to follow when using a plastic material (tarpauline) for drying:

- Find a large plastic sheet or several small plastic sheets or plastic sacks that can be laid out so that they overlap to form a large covered area. Make sure there are no sharp objects on the ground that will tear the plastic.
- Build a flattened mound of hard-packed earth on which to place the sheet. If a level ground is used, dig a shallow trench around the area on which the plastic will be placed to direct any rain water away from the drying floor.
- Put the clean grain on the plastic, in a layer not exceeding 4 cm deep. Stir the grain with hands, a rake or other suitable tool at least every hour making sure all parts of the grain are exposed to the sun. This will aid faster drying.
- As the grain dries, moisture from the grain will collect on the plastic. After the grain has dried for two hours, push all the grain to one half of the plastic sheet. Let the uncovered part of the plastic sheet dry off for 5 minutes or so. Then push all the grain back onto the half of the plastic sheet that is now dry and then let the other half of the plastic sheet dry for 5 minutes, before re-spreading the grain out over it.
- Cover the grain at night. Push all the grain onto one half of the plastic sheet and fold the remaining plastic over it as a cover, or place an extra piece of plastic over the grain. Put boards, planks, rocks, or other heavy things on the corners and edges of the plastic cover to prevent it from being blown off during the night.

The speed of drying largely depends on the weather, how hot and dry it is, and how moist the grain is at the start. If at the start of drying, the

grain holds only 20% moisture content and is exposed to 5 hours of sunny dry weather daily, such grain can achieve 12% moisture content, dry enough for storage in two or three days.



Fig. 21 (a): Drying on cemented floor



Fig. 21 (b): A tarpaulin is a fine drying choice



Fig. 21 (c): Drying on local mats

Threshing

Threshing is the removal of grain from the harvested plant part. The heads are ready for threshing when the grain easily falls off the head with minimal impact. At this stage threshing can be manually done by pounding or beating the heads with sticks or clubs repeatedly until almost all the grains are detached from the heads (Fig. 22). Threshing must be done on a clean hard surface free from soil particles and stones such as a cemented floor or tarpaulin. The straw that remains after threshing may be used as a source for fuel, thatching material for traditional houses, mulch or animal feed.



Fig. 22: A group of youths thresh heads of finger millet

Cleaning

Cleaning refers to separation of contaminants from produce, and complete removal of the contaminants so that the cleaned produce is free from re-contamination.

The contaminants for millets are any foreign material such as sand (soil), small stones, leaves, shrivelled seeds, off-type seeds, broken seeds, glumes, sticks, chaff, parts of stems, insects, animal hair, animal excreta (e.g. rat and insect faeces) and more annoyingly, metal pieces. Cleaning is traditionally done by three techniques namely:

Winnowing:

This is a process of millet cleaning whereby approximately 2 kg to 3 kg of threshed millet grains are placed on a flat reed- or raffia-woven basket (Fig. 23 (a)) and winnowed by up and down strokes. In this process, the basket is jerked up and down so that the grains are thrown up in the air and allowed to fall back onto the basket. The sand and other light contaminants are separated to the front of the basket from where they are thrown off by a jerky motion, or are removed by hand. Often, the light contaminants are blown off by the mouth. The process is labourious, and time consuming, and may take up to a whole day to clean a 100 kg bag of millet.

Aspiration:

This is a traditional cleaning process whereby the difference in densities between the good millet grains and contaminants is exploited to separate the contaminants from millet grains. Slightly windy conditions are required for this method. The wind current blows the light contaminants from the relatively heavy grains (Fig. 23 (b)). About 10 kg of threshed millet grains are placed in a tin or basket and poured from above the head to fall on to the ground usually lined with either a tarpaulin or mat. The wind blows off the chaff leaving a heap of clean grains. This method is much faster than winnowing since up to 4 or 5 bags of 100 kg each may be cleaned in 1 hour. However, the method is not effective in separating sand, stone and metal contaminants which are as heavy as (or even heavier than) the millet grains.

Screening:

This is a process where a sieve or set of sieves (Fig. 23 (c)) is used to separate abnormally small grains from the good ones, the good grain is retained on the sieve while smaller heavier impurities fall through it. Screening also removes sand particles as well as broken pieces of millet grains. A double handed sieve can be operated by two people, who roll it back and forth. The mesh sizes of sieves for millet is usually smaller (< 2mm).



Fig. 23 (a): A woman uses a locally woven basket for cleaning



Fig. 23 (b): Women using wind to their advantage in cleaning millet



Fig. 23 (c): Screen used in the initial sorting phase

Packaging

After threshing, drying and cleaning, millets are usually bagged into 100 kg sisal or plastic bags (Fig. 24) and sealed ready for transportation to distant markets. Sometimes millet grains may be packed in bags sewn from artificial polythene bags for either transportation or storage.

Storage

Storage of finger millet grain is an essential component of the whole production system. It facilitates several farmer objectives, namely, preserving food for the future and avoiding food shortage, providing seed for the next growing season, and allowing sales of grain at a time when the market price is good.

Grain stores are suitable if a moisture content of 10-12% prevails. There are different options for storing finger millet grain but it depends on the intended duration of storage (i.e. short or long term). For short term storage, the grain can be packaged in sacks and sewn/stitched shut. Where grain is stored in bags these should be placed on wooden pallets (or their equivalent) and off the wall to prevent absorption of moisture from the floor and walls. Regular checking of the stored grain for any problem should be undertaken. For long term storage, use of improved grain storage structures is encouraged such as metallic silos is encouraged.



Fig. 24: Produce packed in 100 kg bags

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Table 1(a): Improved Finger Millet Varieties

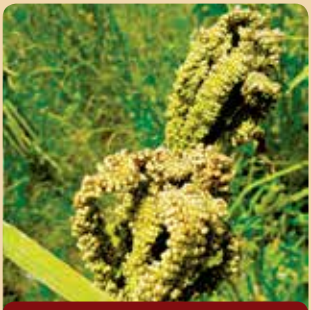
Crop Variety	Crop Characteristics
 **NAROMIL 1**	Yield Potential: 3-4.1 t/ha Duration: 100 days Medium tall, erect and nonlodging plant type. Resistant to leaf blast & moderately resistant to neck and finger blast. 8 – 9 top curved fingers per panicle. 5 - 9 productive tillers. Synchronized maturity, Easy thresh ability. Rich in protein (9.8%) and calcium (0.66%) Highly suitable for value addition.
 **NAROMIL 2**	Yield Potential: 2.93-4.07 t/ha Duration: 102 days. Medium tall, erect and non-lodging plant type. Resistant to leaf blast & moderately resistant to neck and finger blast. Nutritionally rich grain (12.2% protein) Diastatic Unit 45.0 Highly suitable for value addition.
 **NAROMIL 3**	Yield Potential: 2.7 – 4.4 t/ha Duration: 110 days. Medium tall, erect and nonlodging plant type. Resistant to blast. Nutritionally rich grain (10.5% protein) Rich in Iron (88.44 mg/kg). Diastatic Unit 40.0 Highly suitable for value addition.

Table 1(b): Improved Finger Millet Varieties

Crop Variety	Crop Characteristics
 NAROMIL 4	Yield Potential: 2.8 –3.7 t/ha Duration: 85 days Resistant to blast Nutritionally rich grain (7.9% protein) Diastatic Unit 44.0 Excellent malting quality, Highly suitable for value addition
 NAROMIL 5	Yield Potential: 2.6-3.5 t/ha Duration: 100 days. Medium tall, erect and nonlodging plant type. Drought tolerant, Stay green, and Good brewing. Resistant to leaf blast & moderately resistant to neck and finger blast. Nutritionally rich grain (12.2% protein). Diastatic Unit 50.3 Highly suitable for value addition.
 SEREMI 2	Yield potential 1.85-2.5 t/ha Duration 75 days. Tolerant to drought. Grain colour is dark brown. Highly suitable for value addition.



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